

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\mathcal{L} = \frac{1}{g^2} \int \left(-\frac{1}{4} F_{mn} F^{mn} + \frac{1}{2} D_m \Phi^{ab} D^m \Phi_{ab} + \frac{1}{4} \{ \Phi^{ab}, \Phi^{cd} \}_* \{ \Phi_{ab}, \Phi_{cd} \}_* + \right. \\ \left. + i \lambda_a \sigma^m D_m \bar{\lambda}^a + i \{ \lambda_a, \lambda_b \}_* \Phi^{ab} + i \{ \bar{\lambda}^a, \bar{\lambda}^b \}_* \Phi_{ab} \right) .$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\mathbb{E} [\rho_2(Y_2, y_2)] - 1 \leq \mathbb{E} [\mathbf{d}_{S_2}(S_2, s_2)^{p_2}] = \mathbf{W}_{p_2} \left(\gamma_{2,3}^{\eta_1, \delta}, \delta_{s_2} \right)^{p_2} \leq \left[\delta^{1/p_2} + \mathbf{W}_{p_2}(\mu_2, \delta_{s_2}) \right]^{p_2},$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\left\{ \begin{array}{l} \delta^{(1)} A_\mu = i \bar{\epsilon} \sigma_\mu \psi \\ \delta^{(1)} \psi = \frac{i}{2} ([A_\mu, A_\nu] - i \alpha \epsilon_{\mu\nu\lambda} A_\lambda) \sigma^{\mu\nu} \epsilon \end{array} \right. , \quad \left\{ \begin{array}{l} \delta A_\mu^{(2)} = 0 \\ \delta^{(2)} \psi = \xi. \end{array} \right.$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\left\| \sum_{|j|>\ell} \sum_{Q \in Q^j} w(Q) \psi_j(x, x_Q) q_j h(x_Q) \right\|_{\dot{\mathcal{F}}_{p, \mathbb{D}}^{\alpha, q}} \\ = C \left\| \left\{ \sum_{k \in \mathbb{Z}} \sum_{Q \in Q^k} \left(\mathfrak{R}^{k\alpha} \Big|_{q_k} \left(\sum_{|j|>\ell} \sum_{Q \in Q^j} w(Q) \psi_j(\cdot, x_Q) q_j h(x_Q) \right) (x_Q) \right) \right\}^q \chi_Q \right\|_{L_\omega^p}^{1/q},$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\delta_1^{(1)} h_{\alpha\beta} = g(\partial_{\pi\kappa} \partial_{\kappa\nu} \xi_{\alpha\pi} h_{\nu\beta} + \partial_{\kappa\nu} \partial_{\nu\beta} \xi_{\kappa\pi} h_{\alpha\pi} - s \partial_{\alpha\pi} \partial_{\kappa\nu} \xi_{\pi\kappa} h_{\nu\beta}) + (\alpha \Leftrightarrow \beta)$$